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Minimum audible movement angle in the horizontal plane as a function of stimulus frequency and bandwidth, source azimuth, and velocity^{a)}

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Minimum audible movement angles (MAMAs) were measured in the horizontal plane for four normal-hearing adult subjects in a darkened anechoic chamber. On each trial, a single stimulus was presented, and the subject had to say whether it came from a stationary loudspeaker or from a loudspeaker that was moving at a constant angular velocity around him. Thresholds were established by adaptively varying stimulus duration. In experiment 1, MAMAs were measured as a function of center frequency (500–5000 Hz), velocity (10°–180°/s), and direction of motion (left versus right). There was no effect of direction of motion. MAMAs increased with velocity, from an average of 8.8° of arc for a target moving at 10°/s to an average of 20.2° of arc for a target moving at 180°/s. MAMAs were higher for a 3000-Hz tone than for tones of lower or higher frequencies, as has been previously reported [D. R. Perrott and J. Tucker, *J. Acoust. Soc. Am.* **83**, 1522–1527 (1988)]. In experiment 2, minimum audible angles (MAAs) were measured with sequentially presented stationary tone pulses (500–5000 Hz), and were shown to exhibit the same dependence on signal frequency that the MAMAs showed (average MAA at 3000 Hz: 8.4°; average MAA at the other frequencies: 3.4°). In experiment 3, MAMAs and MAAs were measured as a function of stimulus bandwidth (centered at 3000 Hz) and listening azimuth (0° vs 60°). Average MAAs decreased monotonically as stimulus bandwidth increased from 0 Hz to wideband (from 8.4° to 1.2° at 0° azimuth; from 11.3° to 1.5° at 60° azimuth). As in experiment 1, MAMAs increased with stimulus velocity, from values comparable to the MAAs for the slowest-velocity (10°/s) targets to 70° of arc or more in the poorest condition (third-octave band of noise presented at a velocity of 180°/s and an azimuth of 60°). MAMAs obtained in the slower-velocity conditions depended in the same way on stimulus bandwidth and listening azimuth that MAAs depended on these variables. In no case was the MAMA ever smaller than the MAA. It is hypothesized that a minimum integration time is required to achieve optimal performance in a dynamic spatial resolution task. Average estimates of this minimum time based on the current data vary from 336 ms (for targets presented at midline) to 1116 ms (for narrow-band targets presented at 60° azimuth). The relatively large magnitudes of these minimum integration times are consistent with the view of the binaural processor as a sluggish system.

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INTRODUCTION

Detection of the motion of sound sources in the horizontal plane has been the subject of several investigations in recent years. The resulting threshold measure from these investigations is called the minimum audible movement angle (MAMA)—the smallest angular distance a moving sound must traverse in the free field to be just discriminable either from a stationary source or from a source moving in the opposite direction. Harris and Sergeant (1971) were the first to report that for very slowly moving sound sources in the horizontal plane, the MAMA was only slightly larger than the *minimum audible angle* (MAA) as determined

with a pair of sequentially presented *stationary* sound sources. Thus, for a source moving in the horizontal plane at an angular velocity of 2.8°/s, the MAMA was between 2° and 4° of arc for tonal signals of 800 to 6400 Hz. By comparison, the MAA for these signals varies from 1° to 3° (Mills, 1958).

Perrott and Musicant (1977) extended the results of Harris and Sergeant, and showed that as the velocity of a moving source increased, the MAMA increased dramatically. Employing a 500-Hz tonal stimulus, they found that a source moving at 360°/s around the subject in the horizontal plane had to traverse over 20° of arc before it could be just discriminated from a stationary source.

Grantham (1986), employing a simulation technique to produce moving 500-Hz tonal targets, replicated some of the results of the earlier investigations. He found that for suffi-

^{a)} Portions of this research were presented at the 119th Meeting of the Acoustical Society of America [*J. Acoust. Soc. Am. Suppl.* **1** **87**, S64 (1990)].

ciently slow target velocities, the MAMA approached the value of the MAA (2° – 5°); however, as velocity increased, the MAMA showed a linear increase, reaching 20° or greater for a velocity of $360^{\circ}/s$.

A. Velocity, duration, angular extent: Confounded variables

In the movement detection tasks described above, the variables of stimulus velocity, stimulus duration, and angular extent traversed (MAMA) are inevitably confounded, since the last of these is the product of the first two. The nature of the confound may be illustrated by considering an example: In a typical MAMA determination the velocity of a loudspeaker would be held constant (say at $45^{\circ}/s$) and on each trial the subject would have to respond whether a pulsed stimulus came from this moving loudspeaker or from a stationary loudspeaker. Task difficulty would be manipulated by controlling the duration of the pulses: The longer the presentation, the farther the moving sound would travel during its on-time, and the easier the task would be. *Threshold* duration can be specified as that duration for which performance reaches some specified level (e.g., 70.7% correct). If we find in this example that threshold duration is 245 ms, then we compute the MAMA as $45^{\circ}/s \times 245 \text{ ms} = 11.0^{\circ}$ of arc.

In a typical experiment, MAMAs would be thus determined for a number of source velocities. Considering one additional example, if we repeat the experiment for a source moving at $90^{\circ}/s$, we might find that threshold duration is 161 ms, and the MAMA is $90^{\circ}/s \times 161 \text{ ms} = 14.5^{\circ}$ of arc.

In this example, we see that as the velocity of a moving target is increased, the distance that must be traversed to make the discrimination (i.e., the MAMA) increases, but the stimulus duration required to make the discrimination *decreases*. These relationships, together with the example data points just described, are illustrated schematically in the left half of Fig. 1.

Published results indicate that the MAMA grows essentially linearly with velocity, as schematized in Fig. 1(a) and represented in the following equation:

$$\text{MAMA} = Tv + m, \quad (1)$$

where v is the angular velocity, and T and m are fitting parameters ($m = 7.5^{\circ}$ and $T = 78 \text{ ms}$ in the example we are considering). A straightforward interpretation of this relationship is that the y -intercept (m) represents the MAA; that is, the MAA is a special case of the MAMA, representing the end point of the velocity continuum, where $v = 0^{\circ}/s$.

To represent threshold stimulus *duration* as a function of velocity, we divide Eq. (1) by v :

$$d_t = T + m/v, \quad (2)$$

where d_t is threshold duration (that is, MAMA/v). This relationship [illustrated in Fig. 1(b)] implies that threshold duration for a moving target is comprised of two parts: the time it takes the target to traverse the MAA (m/v) and some constant time T . Threshold duration asymptotes to this constant T as velocity increases. In some sense, then, the constant T , which is the slope of the function in Fig. 1(a), repre-

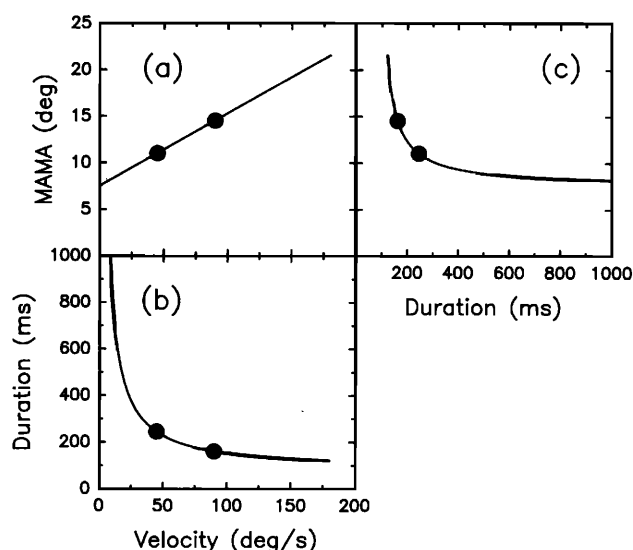


FIG. 1. Schematic functions with two example data points, illustrating the relationships among MAMA, stimulus velocity, and stimulus duration. (a) MAMA as a linear function of stimulus velocity; (b) threshold stimulus duration as a function of stimulus velocity; (c) MAMA as a function of threshold stimulus duration. See text.

sents a minimum time required for processing moving targets.

There is clearly a trade between the MAMA and threshold duration: As velocity increases, a shorter and shorter time is required to make the discrimination [Fig. 1(b)], but at a "cost" of an increasingly large angle that must be traversed [Fig. 1(a)]. The implication is that at the higher velocities performance is governed primarily by temporal factors [i.e., T dominates in Eq. (2)], and that the distance traversed is not a salient cue. However, at the lower velocities, spatial resolution is the governing factor [m assumes increasing weight in Eq. (2)], and distance traversed is the primary cue.

The increasing salience of spatial resolution as velocity is decreased may be appreciated by plotting the variables in yet another way: in Fig. 1(c), the MAMA is plotted as a function of stimulus duration. This relationship is derived by replacing v in Eq. (1) with MAMA/d_t , which results in the following relation:

$$\text{MAMA} = m/(1 - T/d_t). \quad (3)$$

On this axis system, velocity is represented by the slope of the line through any given point and the origin. This method of plotting thresholds reveals that as stimulus duration is increased, MAMA decreases monotonically, approaching an asymptote (a value equal to or close to the MAA— 7.5° in the example). Plotting the data in this way, as was done in Grantham (1986), allows an estimate of the minimum integration time (MIT) required for *spatial resolution performance to reach an optimum level*. This minimum time differs from T , discussed above, in that it represents the time required for *optimum* performance and is based on the pattern of improved performance as velocity *decreases*, while the time constant T represents the minimum time required to

perform the task as velocity *increases*, irrespective of the magnitude of the MAMA. We believe the more interesting index in the present application is the MIT, although for a given criterion level of spatial resolution, it can be shown to be a simple multiple of the time constant T .

Based on the pattern of the data plotted in the manner as schematized in Fig. 1(c), Grantham (1986) estimated the MIT for horizontal motion detection to be on the order of 150–300 ms. That is, for velocity conditions that enabled at least this much of a temporal window, MAMAs had practically reached their asymptotic limit (close to the MAA), while velocity conditions that resulted in threshold durations less than this MIT resulted in increasingly large MAMAs.

B. Comparison of MAMAs and MAAs

To the extent that the MAA can be considered a special (zero-velocity) case of the MAMA, the relationship represented by Eq. (1) and plotted in Fig. 1(a) implies that the MAA should be similar in magnitude to MAMAs determined with low-velocity targets. Perrott and Tucker (1988) measured MAMAs as a function of stimulus frequency and velocity and compared their results to the MAAs reported by Mills (1958). They found that the MAMA's dependence on stimulus frequency paralleled very closely the MAA's dependence on frequency: Best performance was for stimuli between 500 and 1000 Hz, performance deteriorated until it reached its worst level for frequencies between 1500 and 2000 Hz, then it improved slightly as frequency was further increased to 3700 Hz. MAMAs at all frequencies increased with velocity (replicating results from Perrott and Muscant, 1977, and Grantham, 1986). Furthermore, the MAAs reported by Mills were about half the value of the best MAMAs reported by Perrott and Tucker (their slowest velocity was 8°/s); thus, it appears that Mills's MAAs at each frequency were *smaller* than the y intercepts of Perrott and Tucker's data plotted on the coordinate system of Fig. 1(a).

The purpose of the first two experiments of the current paper is to extend the results of Perrott and Tucker (1988) and to determine, by comparison of the MAMA and MAA from the same subjects, whether the MAMA for pure tones exhibits the same frequency dependence that the MAA does. The third experiment examines the effects of stimulus bandwidth and listening azimuth on MAMAs and MAAs. For each of these manipulations, we were particularly interested in whether the MAA could be interpreted as a "0°/s MAMA;" stated differently, we wanted to test the hypothesis that under the variety of conditions we employed, the statically measured MAA represents an upper limit of auditory spatial resolution in the horizontal plane.

I. EXPERIMENT 1

In the first experiment MAMAs were determined for pure-tone stimuli as a function of stimulus frequency, velocity, and direction of motion.

A. Subjects

Three females and one male (TT), ages 22–24, with normal hearing and normal word recognition ability (ANSI,

1969) were paid subjects in this experiment. None had had previous experience in psychoacoustic tasks. Each was tested 2–3 times per week in sessions that lasted about 2 h; frequent rest intervals were provided. For each subject, practice was provided until performance had stabilized prior to formal data collection (typically 2–5 sessions).

B. Apparatus and stimuli

Subjects were tested individually in a darkened anechoic chamber. During a session, the subject was seated in the center of the room and instructed to maintain a steady upright, forward orientation during the course of the session without tilting the head either to the side or forward. No bite-bar was employed.

As shown in Fig. 2, 13 stationary loudspeakers (JBL-8110H high compliance, full range) were positioned in a horizontal arc at ear level 1.8 m in front of the listener (in experiment 1, only the two center loudspeakers of this array, $\pm 1.8^\circ$ from midline, were employed). Just inside the arc of the stationary loudspeakers, two additional loudspeakers were suspended from opposite ends of an overhead boom (see Fig. 2). These loudspeakers could move 360° around the listener at the same height as the stationary speakers.

Stimuli were pure tones generated by a General Radio Model 1310-B Oscillator (frequencies: 500, 1000, 3000, or 5000 Hz), whose level was set to a constant 0.316 V at the input to the loudspeakers. The resulting nominal levels measured at the position of the subject's head were 50, 58, 65, and 65 dBA, respectively. To prevent subjects from learning characteristics of individual loudspeakers, a random amount of attenuation (0.0–5.0 dB) was applied to each stimulus prior to its presentation.¹

Stimulus velocities of 10, 20, 45, 90, and 180° /s, in both leftward and rightward directions, were employed [however, one of the four subjects (AC) was tested with only rightward moving targets].

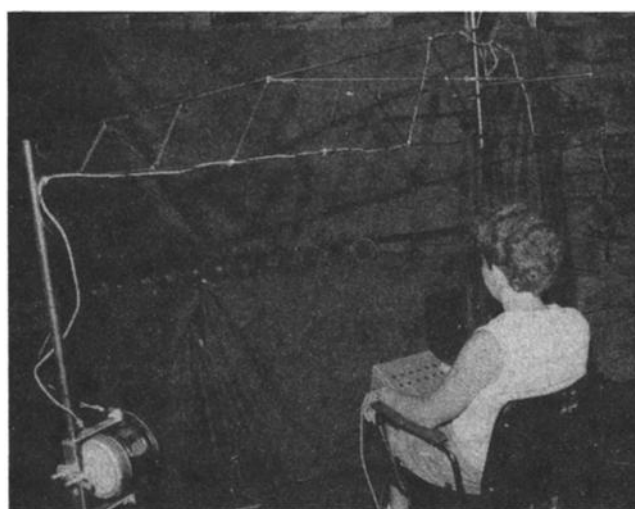


FIG. 2. Arrangement of loudspeakers in the anechoic chamber. Moving targets were presented through one of the two loudspeakers suspended from the ends of the overhead boom, which moved around a pivot point directly over the subject's head. Stationary targets were presented from one of the loudspeakers in the fixed array.

C. Procedure

During a single MAMA determination, stimulus frequency, velocity, and direction were held constant. Prior to the initiation of the trial sequence, the boom was accelerated to the desired velocity and maintained at this velocity throughout the MAMA determination, which typically lasted 3–15 min (depending on the velocity).² A single-interval, forced-choice procedure was employed: on each trial when one of the two moving speakers reached an azimuth near 0° (directly in front of the subject), a single stimulus was presented. On half of the trials the stimulus came from one of the two stationary loudspeakers situated at $\pm 1.8^\circ$, and on the other half, it came from the moving loudspeaker that was just traversing midline. The subject indicated by buttonpress his/her judgment of whether the stimulus appeared to be moving or stationary. Immediate feedback was presented by lights on the button box.

To prevent subjects from responding based on onset (or offset) positions of the stimuli, stimulus onsets from the moving loudspeakers were jittered in time from trial to trial such that the moving stimulus's end point (for "moving" trials) occurred at a random position within $\pm 1.8^\circ$ of midline.³ Similarly, for "stationary" trials, the stimulus came at random either from the loudspeaker at $+1.8^\circ$ azimuth, or from the loudspeaker at -1.8° azimuth. All stimuli were gated by a Coulbourn S84-04 rise-fall gate (10-ms, quarter-cycle sine rise-decay times) without regard to the instantaneous phase of the tone. To mask any switching transients that occurred between trials (when the loudspeaker for the next trial was being selected), a low-level (30 dBA) broadband noise was present between trials. This background noise was gated off at least 333 ms prior to the subsequent stimulus presentation.

Threshold was determined by adaptively varying stimulus duration from trial to trial (whether the target was stationary or moving) using a two-down, one-up tracking procedure (Levitt, 1971).⁴ A threshold duration corresponding to 70.7% correct was thus obtained (average of the last eight of ten total reversals) and then converted to the angular distance traversed ($\text{MAMA} = \text{threshold duration} \times \text{velocity}$). After a threshold determination was completed, there was about a 1-min rest before the next sequence began (typically involving a different velocity and frequency).

A MAMA was accepted only if the standard deviation of the reversals from which it was computed was less than or equal to one-third of its threshold value (thus, thresholds based on large tracking excursions were rejected). At least three acceptable thresholds for each subject and for each condition were obtained on at least two different days. If the standard deviation across these three determinations was greater than one-third of the mean, additional thresholds were obtained, either until the standard deviation was less than one-third the mean, or until six thresholds were obtained. Data reported here for each condition are thus based on the means of 3–6 threshold determinations.

D. Results and discussion

Results from experiment 1 are displayed in Fig. 3: MAMAs, averaged across the four subjects, are plotted as a func-

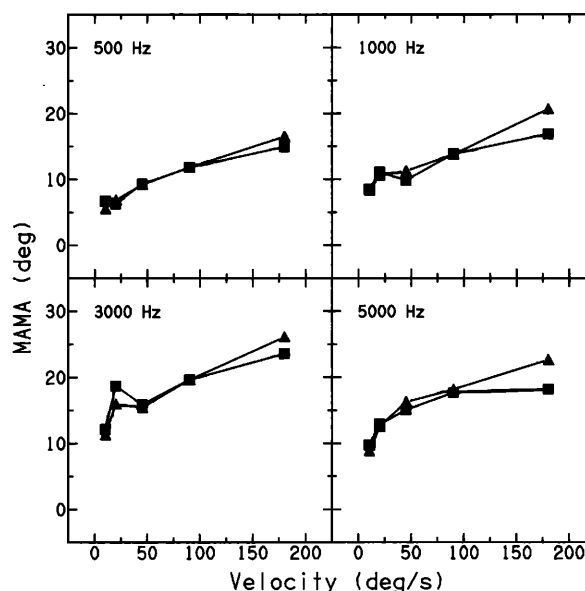


FIG. 3. Average MAMAs across the four subjects as a function of target velocity. Each panel contains the data for a different frequency of the pure-tone stimulus. $\blacktriangle$: Rightward moving targets. $\blacksquare$: Leftward moving targets.

tion of velocity, with direction of motion as the parameter. Data from each frequency condition are plotted in a separate panel. A three-factor, repeated-measures ANOVA (5 velocities $\times$ 4 frequencies $\times$ 2 directions) indicated that none of the interactions among the three main factors was significant ($p > 0.05$).

1. Effect of direction of motion

The main effect of direction of motion was not significant ($F = 0.05$; $df = 1, 2$; $p > 0.8$). All subsequently presented data will be averaged across direction.

2. Effect of velocity

The main effect of velocity was highly significant ($F = 32.91$; $df = 4, 8$; $p = 0.0001$). Figure 3 shows that the MAMA increased as a function of stimulus velocity for all frequencies tested. Averaged across subjects, directions, and frequencies, the MAMA at $10^\circ/\text{s}$ was 8.8° , while the average MAMA at $180^\circ/\text{s}$ was 20.2° . This result, which replicates previous findings (Perrott and Musicant, 1977; Grantham, 1985, 1986; Perrott and Tucker, 1988), indicates that, as source velocity is increased, a greater and greater angular extent is required in order to discriminate the moving target from a stationary one. As noted in the introduction, this finding is consistent with the idea that there is a minimum integration time required for the auditory system to perform optimally in a spatial resolution task.

3. Estimate of minimum integration time for spatial resolution

An estimate of the minimum integration time (MIT) for each condition was made from the velocity functions in the following manner: For a given subject and frequency, the MAMAs (averaged across direction) were plotted as a func-

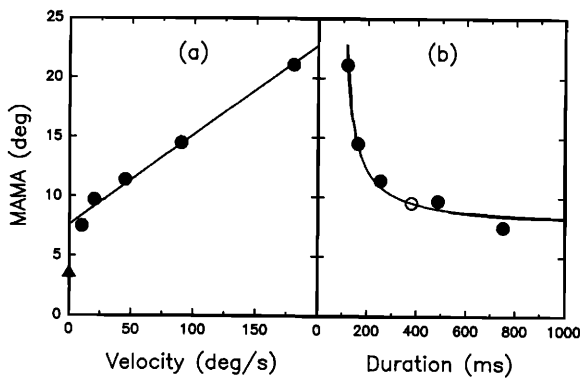


FIG. 4. (a) MAMAs as a function of velocity in one condition from experiment 1 (subject TT, 1000-Hz tonal stimulus). The straight line is the linear regression of the MAMAs (closed circles). The MAA from experiment 2 for this stimulus and subject is represented by the closed triangle at velocity = 0°/s (not included in the regression). (b) The same MAMAs are replotted as a function of duration, where duration = MAMA/velocity. Thus these durations represent the stimulus on-times required for discrimination. The linear regression from panel (a) is also replotted as a function of duration, resulting in an asymptotically decaying function. The open circle is the point on this fitted function that is 25% greater than the asymptote; the corresponding duration is taken as an estimate of the subject's minimum integration time for this condition (379 ms in this example). See text.

tion of velocity, and a linear regression was performed as illustrated Fig. 4(a). The same data points, as well as the linear fit, were then replotted as a function of *duration*, as illustrated in Fig. 4(b) (i.e., duration = MAMA/velocity). The latter figure, derived as explained in the introduction and schematized in Fig. 1(c), shows clearly the asymptotic behavior of the MAMA functions as duration increases. Arbitrarily picking a point which is 25% greater than the asymptote [i.e., 25% greater than the *y*-intercept in Fig. 4(a)], we designated the duration corresponding to this point on the fitted function as the MIT. This duration represents the point at which the MAMA has *almost* reached its asymptotic value. In the illustration, the MIT is 379 ms [indicated by the open circle in Fig. 4(b)].⁵

Averaged across subjects, the MITs estimated in this manner were 292, 294, 335, and 295 ms, for 500-, 1000-, 3000-, and 5000-Hz targets, respectively (individual estimates are included in Table I). Thus, despite significant differences in auditory spatial resolution based on stimulus frequency (see below), the minimum time required for optimum MAMAs is about the same for all frequencies tested.

The relatively large values estimated for MIT in this experiment are consistent with the fact that the binaural system responds quite sluggishly to dynamic variation of interaural cues (e.g., Blauert, 1968, 1972; Grantham and Wightman, 1979; Grantham, 1984; Kollmeier and Gilkey, 1990). Thus it is probable that the minimum time required to perform optimally in a spatial resolution task is based on limitations in the binaural system's response to dynamic stimuli.

4. Effect of frequency

The main effect of frequency was significant ($F = 8.45$; $df = 3, 6$; $p = 0.01$). To better examine the effects of stimu-

lus frequency, the data from Fig. 3 are replotted in Fig. 5, which displays MAMAs as a function of frequency, with velocity as the parameter. Here, data have been averaged across both subjects and direction. Individual data, averaged across direction, are also presented in Table I (columns 2–6, representing velocities of 10°–180°/s, and rows 1–4 under each subject, representing the tonal stimuli 500–5000 Hz). Mean data and standard deviations across subjects are also included in Table I.

Figure 5 shows that as frequency increased from 500 to 5000 Hz, MAMAs at all velocities first increased, reaching a maximum at 3000 Hz, then decreased slightly at 5000 Hz. Newman-Keuls *post-hoc* analysis revealed that MAMAs at 3000 Hz were significantly greater than MAMAs at the other three frequencies ($p < 0.05$), which were not significantly different from each other. These data replicate those of Perrott and Tucker (1988), who found a similar pattern of results for frequencies from 500 to 3700 Hz. These latter investigators also pointed out that although the MAAs reported by Mills in 1958 were about two times lower than Perrott and Tucker's lowest MAMAs (obtained with targets moving at 8°/s), the *shape* of the two functions was practically identical.

In order to provide a more straightforward comparison between spatial resolution in static versus dynamic conditions, we measured MAAs for our subjects in the next experiment. With these measurements, we can compare MAAs and MAMAs from the same subjects in the same laboratory, employing a procedure that allows an analogous response set for the two tasks.

II. EXPERIMENT 2

MAAs were determined for three of the four subjects tested in experiment 1 (the fourth was unavailable). The anechoic chamber, the testing conditions, and the stimuli (tones of 500, 1000, 3000, or 5000 Hz) were the same as had been employed in experiment 1.

A. Procedure

Only the stationary loudspeakers in the horizontal array (see Fig. 2) were employed in this experiment. In a given block of 50 trials, two designated loudspeakers from this array were employed. On each trial two sequential tone bursts were presented to the subject: The first burst always came from the leftmost of the designated pair of loudspeakers; the second burst came either from the same loudspeaker, or from the other (rightmost) loudspeaker. As in experiment 1, the intensity of the presentation occurring in each interval was randomly varied over a range of 5.0 dB to keep subjects from performing based on individual loudspeaker characteristics. The subject had to respond whether the second burst appeared to come from the same location as the first, or from a location to the right of the first. Immediate feedback was provided.

This "same-different" task was chosen because it paralleled the response set required in the MAMA task: in both cases, the subject had to judge whether the ending location was the same as the first (i.e., a stationary stimulus in the

TABLE I. Average thresholds (MAMAs) in degrees of arc as a function of velocity and stimulus type. Thresholds were obtained at a 0° listening azimuth. Standard deviations are given in parentheses. For MAAs (column 1), thresholds were interpolated (or extrapolated^a) from fitted psychometric functions, and the standard errors of the estimate (when available) are given in parentheses. Estimated minimum integration times are given in the last column. See text.

	Velocity (°/s)						Minimum integration time (ms)
	0 (MAA)	10	20	45	90	180	
Subject: TT							
500 Hz	1.0 ^b	4.7 (1.5)	5.0 (1.0)	8.1 (0.8)	10.6 (1.6)	14.7 (4.5)	296
1000 Hz	1.7 ^b	7.5 (2.6)	9.7 (2.5)	11.4 (3.2)	14.5 (3.0)	21.1 (4.7)	379
3000 Hz	3.6 (4%)	11.9 (0.8)	18.3 (4.0)	16.6 (2.2)	21.7 (4.7)	28.1 (5.0)	410
5000 Hz	4.4 (3%)	7.2 (2.3)	10.9 (2.6)	11.4 (2.2)	12.8 (3.0)	17.2 (3.8)	246
Third octave	2.3 ^a (2%)	7.2 (0.2)	10.2 (2.4)	10.3 (1.1)	15.2 (1.1)	23.1 (1.0)	445
One octave	1.1 ^a (2%)	7.2 (0.8)	8.8 (0.9)	11.3 (1.1)	16.1 (3.8)	15.4 (0.5)	239
Wide band	1.3 ^b	5.4 (0.4)	5.2 (0.5)	9.3 (1.6)	9.9 (1.4)	19.2 (2.7)	401
Subject: MW							
500 Hz	4.2 (9%)	7.5 (1.4)	8.0 (1.9)	10.8 (2.9)	15.2 (2.6)	20.9 (3.7)	402
1000 Hz	3.6 (9%)	9.2 (1.4)	9.4 (2.4)	10.6 (1.6)	15.2 (3.6)	19.6 (3.9)	322
3000 Hz	12.2 (11%)	11.7 (2.7)	14.6 (3.0)	14.1 (3.1)	17.6 (3.6)	25.5 (4.9)	379
5000 Hz	4.7 (5%)	9.0 (2.2)	13.2 (2.8)	17.5 (5.5)	19.3 (5.2)	22.0 (6.3)	325
Third octave	1.3 ^b	9.1 (0.4)	9.2 (0.9)	11.5 (0.7)	15.0 (1.7)	20.8 (1.7)	354
One octave	1.7 ^a (1%)	5.8 (0.1)	6.8 (1.1)	11.5 (0.9)	11.2 (2.3)	17.6 (3.3)	322
Wide band	1.0 ^b	5.2 (1.1)	5.8 (1.7)	6.4 (5.4)	11.6 (1.4)	16.0 (4.1)	332
Subject: AG							
500 Hz	—	5.3 (0.9)	6.5 (1.5)	8.0 (2.6)	9.2 (2.4)	9.7 (4.1)	116
1000 Hz	—	9.1 (2.1)	13.3 (4.1)	8.6 (3.7)	11.8 (4.4)	16.3 (5.8)	172
3000 Hz	—	11.5 (1.1)	19.1 (3.7)	16.2 (3.0)	19.6 (4.1)	22.7 (5.0)	241
5000 Hz	—	10.4 (2.2)	13.5 (2.4)	16.1 (3.9)	19.4 (2.2)	19.9 (6.3)	248
Third octave	—	8.8 (0.9)	11.3 (1.7)	13.8 (4.4)	21.5 (1.3)	26.0 (4.9)	503
One octave	—	6.5 (0.4)	6.5 (0.8)	11.3 (0.7)	15.2 (1.1)	17.9 (2.2)	347
Wide band	—	5.3 (0.2)	6.2 (1.1)	8.3 (1.8)	12.1 (3.0)	16.4 (4.1)	327
Subject: AC							
500 Hz	3.3 ^a (1%)	6.0 (0.7)	7.3 (1.9)	9.1 (1.3)	11.9 (0.8)	18.4 (3.1)	354
1000 Hz	2.0 ^b	6.9 (1.2)	9.0 (2.1)	10.8 (1.3)	10.7 (0.6)	18.5 (4.5)	305
3000 Hz	9.3 (4%)	11.8 (0.5)	16.5 (3.6)	15.5 (4.0)	18.7 (2.3)	24.1 (5.1)	310
5000 Hz	6.0 (4%)	11.5 (2.2)	13.0 (1.7)	18.3 (4.7)	20.9 (1.8)	24.3 (1.7)	359
Third octave	3.4 ^a (3%)	9.2 (0.3)	10.2 (1.6)	11.3 (0.9)	17.2 (3.1)	21.7 (4.8)	379

TABLE I. (Continued.)

	0 (MAA)	10	20	Velocity ($^{\circ}$ /s)			Minimum integration time (ms)
				45	90	180	
One octave	4.1 (6%)	7.2 (0.9)	9.1 (0.8)	11.2 (1.2)	14.0 (2.7)	24.9 (2.2)	499
Wide band	1.4 ^b	5.0 (1.5)	5.6 (0.8)	8.7 (0.5)	14.4 (2.4)	17.7 (2.4)	387
Mean across subjects							
500 Hz	2.8 (1.7)	5.9 (1.2)	6.7 (1.3)	9.0 (1.3)	11.7 (2.6)	15.9 (4.8)	292 (125)
1000 Hz	2.4 (1.0)	8.2 (1.2)	10.4 (2.0)	10.3 (1.2)	13.0 (2.2)	18.9 (2.0)	294 (88)
3000 Hz	8.4 (4.4)	11.7 (0.2)	17.1 (2.0)	15.6 (1.1)	19.4 (1.7)	25.1 (2.3)	335 (75)
5000 Hz	5.0 (0.9)	9.5 (1.9)	12.7 (1.2)	15.8 (3.1)	18.1 (3.6)	20.9 (3.0)	295 (57)
Third octave	2.3 (1.1)	8.6 (0.9)	10.2 (0.9)	11.7 (1.5)	17.2 (3.0)	22.9 (2.3)	420 (67)
One octave	2.3 (1.6)	6.7 (0.7)	7.8 (1.4)	11.3 (0.1)	14.2 (2.1)	19.0 (4.1)	352 (108)
Wide band	1.2 (0.2)	5.2 (0.2)	5.7 (0.4)	8.2 (1.3)	12.0 (1.8)	17.3 (1.4)	362 (38)

^a Extrapolated.^b See footnote 6.

MAMA task) or whether it was to the right of the first (i.e., a moving stimulus in the MAMA task). In both cases, the subject knew in advance the direction of motion (or position change) the stimulus may take (rightward, in this case).

The purpose of this experiment was to measure the MAA under as near optimal conditions as possible. Since the magnitude of the MAA depends on the temporal parameters employed (MAA increases as the interstimulus interval

(ISI) decreases, and to a lesser extent as stimulus duration decreases (Grantham, 1985; Perrott and Pacheco, 1989), we selected durations to insure optimal performance. Accordingly, each stimulus was pulsed on for 300 ms, and the ISI was also 300 ms.

Angular separations of loudspeakers could vary from 3.6° to 25.7° , in approximate multiples of 3.6° . For each condition tested, 2–4 loudspeaker separations were chosen, and a psychometric function was constructed by measuring percent correct performance in at least eight 50-trial blocks (obtained on at least two different days) distributed among the given separations. Threshold was estimated by finding the best-fitting Gaussian ogive to the data and determining from this function the angular separation corresponding to a performance level of 70.7% (thus enabling a direct comparison to the thresholds obtained in experiment 1 obtained with the adaptive two-down, one-up procedure). This value, obtained for each of the four stimuli, is reported here as the MAA.

B. Results and discussion

The obtained MAAs are reported in Table I, column 1, for the three subjects and the four frequencies tested (first four rows under each subject). Also reported, in parentheses, are the standard errors of estimate, which provide a measure of the goodness of fit of the Gaussian ogive employed to fit the psychometric functions. In three cases (for one subject at 500 Hz and two subjects at 1000 Hz), performance was too good at the smallest separation employed (3.6°) to

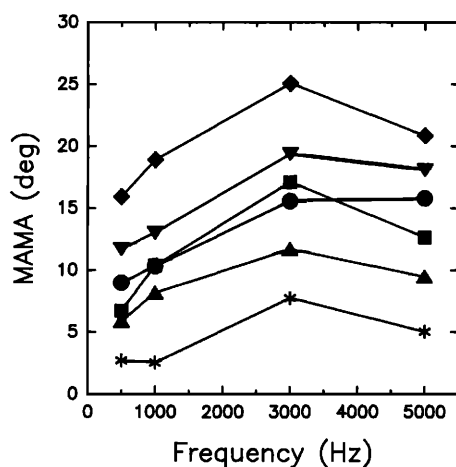


FIG. 5. Average MAMAs across the four subjects and across direction of motion as a function of the frequency of the pure-tone stimulus. Stimulus velocity is the parameter. $\blacktriangle$: 10° /s; $\blacksquare$: 20° /s; $\bullet$: 45° /s; $\blacktriangledown$: 90° /s; $\blacklozenge$: 180° /s. MAAs from experiment 2 are plotted as asterisks (*).

obtain a threshold estimate by *interpolating* from a psychometric function; in these cases, thresholds were estimated either by *extrapolation* from the best-fitting psychometric function, or, if this procedure led to unrealistic estimates, by an estimation based on the subject's performance at the 3.6° separation.⁶

The average MAAs across subjects are plotted in Fig. 5 as asterisks. As can be seen from the figure, the pattern of MAAs parallels that of the MAMAs: Best performance is at 500 and 1000 Hz (average MAA: 2.6°), worst performance at 3000 Hz (average: 8.4°), with a slight improvement again as frequency increases to 5000 Hz (average: 5.0°). The fact that our thresholds are worse by about a factor of two than those reported by Mills (1958) can probably be attributed to the different procedures employed: In our same-different procedure, subjects had to say whether the second pulse was at the same position or was to the right of the first pulse; in the procedure employed by Mills, the second pulse could be to the right or to the left of an initially presented standard pulse. The latter procedure would be expected to yield lower thresholds than the former (see Hartmann and Rakerd, 1989).

Our data replicate and support the conclusions of Perrott and Tucker (1988): The functional dependence of auditory spatial resolution on stimulus frequency is the same in dynamic and static situations. Furthermore, there are no cases in which the dynamic measure of resolution is ever *better* than the static measure. In fact, in all cases, the MAA was less than the *y* intercept of the linear fit to the MAMA-by-velocity function [by 53% on average—see Fig. 4(a)]. This result suggests that the interpretation of the MAA as a “zero-velocity” MAMA may be an oversimplification. However, it appears that under the conditions tested here, the MAA defines the limit of auditory spatial resolution: Dynamic measures of spatial resolution might approach, but do not reach the level of the optimum MAA.

III. STATIC VERSUS DYNAMIC SPATIAL RESOLUTION: STIMULUS BANDWIDTH

Perrott and Marlborough (1989) have reported some results that suggest, at first glance, that the MAMA can be *smaller* than the MAA. Using pink noise stimuli presented at a velocity of 20°/s, they measured MAMAs of about 1° of arc—considerably lower than the MAMAs we measured at 20°/s (average MAMA: 6.7° at 500 Hz). One likely source of the difference in obtained MAMAs between the two studies is the use by Perrott and Marlborough of a wideband stimulus, which would be expected to lead to improved performance compared to that obtained with pure tones.

Perrott and Marlborough compared the MAMAs obtained in their “standard” condition to minimum audible angles obtained under the same conditions except the loudspeaker, instead of remaining activated throughout its trajectory, was activated only at the onset and offset of its trajectory. Thresholds in the latter case, in which the moving targets were “marked” by 10-ms pulses, were about 50% higher than thresholds in the standard, “filled” case. The authors interpreted these results as arguing against the notion that the discrimination of motion can be reduced to a

simple comparison of the location of a source's end points; rather, these authors argued that motion of auditory targets may be appreciated directly, without requiring prior analysis of spatial and temporal differences.

The results from Perrott and Marlborough (1989) contrast with data from a similar experiment by Grantham (1985) which reported that minimum audible angles were no different in a “marked” condition than in a “filled” condition. The differences between the two studies may be due to the different stimuli employed (the latter study used 500-Hz pure tones, while the former used wideband noise). If the auditory system is able to perform some kind of special processing of moving targets in order to spatially resolve them, it is possible that some minimum bandwidth of the target would be required before the effects of such special processing could be measured.

The MAMAs reported by Perrott and Marlborough are indeed impressively low. In an attempt to replicate this extremely good dynamic performance and in order to better understand the role of stimulus bandwidth in comparing dynamic and static spatial resolution, experiment 3 measured MAMAs and MAAs as a function of bandwidth. In addition, experiment 3 introduced the variable of listening azimuth: MAMAs and MAAs were measured at both 0° and 60° azimuth.

IV. EXPERIMENT 3

MAMAs and MAAs were obtained for the same four subjects who were employed in experiment 1 (however, we obtained no MAAs from subject AG). Except for the stimuli, all aspects of the testing environment and procedure were identical to those described for experiment 1 (MAMA) and experiment 2 (MAA).

Three stimuli were employed: (1) a one-third octave band of noise centered at 3000 Hz (2700–3400 Hz); (2) a one-octave band of noise centered at 3000 Hz (2100–4200 Hz); and (3) a wideband noise (500–10000 Hz). Noise was produced by a Coulbourn Gaussian noise generator led through two Kemo programmable filters (low-pass filter skirt was 135 dB/oct, and high-pass filter skirt was 90 dB/oct). Levels of the noise measured at the position of the subject's head were 64, 65, and 64 dBA, respectively. As described above, a random amount of attenuation (0.0–5.0 dB) was introduced prior to each stimulus's presentation, in order to discourage subjects from attending to any subtle loudspeaker differences that may have been present.

Two listening azimuths were employed. For the 0°-azimuth condition, as described for experiments 1 and 2, the subject was oriented in a forward direction (toward the center of the fixed array). For the 60°-azimuth condition, the stimuli were presented from the same loudspeakers as previously described; however, in this case the subject's chair was rotated to the right, and the subject was instructed to orient, before the lights were extinguished, toward a (nonilluminated) visual marker that was placed on the wall at a position 60° from midline. Thus stimulation was from a position 60° to the subject's *left*. In order to be able to compare the noise data to tonal data, MAMAs and MAAs were also

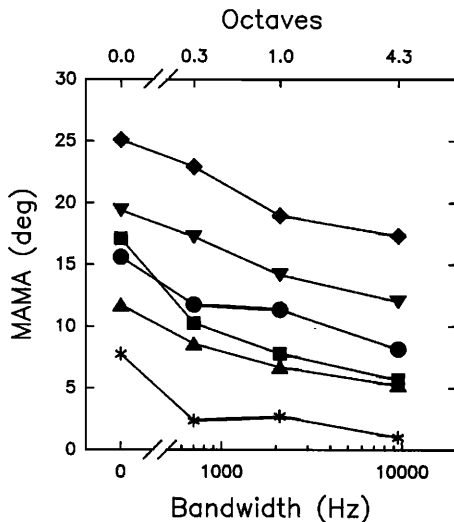


FIG. 6. Average MAMAs across the four subjects as a function of stimulus bandwidth. Listening azimuth is 0° . The data points plotted at the 0-Hz bandwidth are from experiment 1 (the 3000-Hz data points from Fig. 4). Stimulus velocity is the parameter. $\blacktriangle$: $10^\circ/\text{s}$; $\blacksquare$: $20^\circ/\text{s}$; $\bullet$: $45^\circ/\text{s}$; $\blacktriangledown$: $90^\circ/\text{s}$; $\blacklozenge$: $180^\circ/\text{s}$. MAAs are plotted as asterisks (*).

measured for a 3000-Hz tone at the 60° azimuth listening condition.

For the MAMA task target velocities of 10° , 20° , 45° , 90° , and $180^\circ/\text{s}$ (rightward direction only) were employed.

A. Results and discussion

Mean thresholds across subjects are plotted in Fig. 6 (0° azimuth) and Fig. 7 (60° azimuth) as a function of stimulus bandwidth, with velocity as the parameter. The data plotted at 0-Hz bandwidth in Fig. 6 are the 3000-Hz thresholds obtained in experiments 1 and 2. MAAs are plotted in both Figs. 6 and 7 as asterisks.

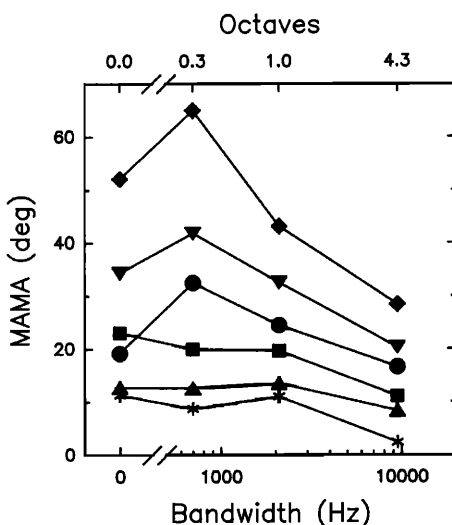


FIG. 7. Average MAMAs across the four subjects as a function of stimulus bandwidth. Listening azimuth is 60° . Stimulus velocity is the parameter. $\blacktriangle$: $10^\circ/\text{s}$; $\blacksquare$: $20^\circ/\text{s}$; $\bullet$: $45^\circ/\text{s}$; $\blacktriangledown$: $90^\circ/\text{s}$; $\blacklozenge$: $180^\circ/\text{s}$. MAAs are plotted as asterisks (*).

Individual and mean data are given in Table I for the 0° azimuth conditions (last three rows under each subject's section) and Table II for the 60° azimuth conditions. In several cases performance was too good at the smallest angular separation employed (3.6°) to enable a threshold estimate by interpolating from the obtained psychometric function; in these cases (noted in the tables) the MAAs were estimated in an alternate manner.⁶

A three-factor, repeated-measures ANOVA (5 velocities $\times$ 4 bandwidths $\times$ 2 azimuths) performed on the results from experiment 3 revealed a significant three-way interaction ($p < 0.002$). Hence, the data were analyzed separately for the two azimuth conditions.

1. Thresholds at 0° azimuth

A two-factor ANOVA (5 velocities $\times$ 4 bandwidths) performed on the MAMAs at 0° azimuth revealed no significant interaction ($p > 0.2$), but highly significant main effects of velocity ($F = 210.24$; $df = 4, 12$; $p < 0.0001$) and bandwidth ($F = 33.01$; $df = 3, 9$; $p < 0.0001$). Figure 6 shows that thresholds increased as velocity increased, as has been observed in all previous MAMA experiments. Furthermore, thresholds decreased monotonically as bandwidth increased for all velocities tested. Thus the auditory system is able to take advantage of signal bandwidth in making spatial resolution judgments. This pattern is also seen for the MAA function (asterisks). Thus, at 0° listening azimuth, the dependence of auditory spatial resolution on stimulus bandwidth is the same for static (MAA) and dynamic (MAMA) stimuli.

2. Thresholds at 60° azimuth

Figure 7 indicates that thresholds at 60° azimuth were, in general, 1.5 to 3 times larger than those at 0° , reaching in some cases 60° of arc or greater. Inter- and intrasubject variability was also greater at 60° than at 0° . Poorer auditory spatial resolution off midline than on midline has been previously demonstrated for both static (Mills, 1958) and dynamic (Grantham, 1986) conditions, and can be understood in terms of the change in interaural cues available in these stimulus conditions (Mills, 1960).

A two-factor ANOVA performed on the 60° azimuth data revealed a significant interaction between velocity and bandwidth ($F = 8.93$; $df = 12, 36$; $p < 0.001$), as well as significant main effects of velocity ($F = 99.97$; $df = 4, 12$; $p < 0.0001$) and bandwidth ($F = 27.60$; $df = 3, 9$; $p < 0.0002$). As can be observed from the figure, MAMAs showed a generally monotonic increase with increasing velocity, although the rate of the increase depended on the bandwidth. We will return to this point below.

In terms of bandwidth, two separate patterns are seen in the data, depending on stimulus velocity. For the three highest velocities, MAMAs showed a surprising *nonmonotonic* pattern as a function of bandwidth: Performance improved sharply as bandwidth increased from third-octave to wide-band, but thresholds were *lower* for the pure tone than for the third-octave band stimulus. For the two slowest velocities ($10^\circ/\text{s}$ and $20^\circ/\text{s}$), on the other hand, MAMAs showed a gradual, monotonic decrease with bandwidth.

TABLE II. Average thresholds (MAMAs) in degrees of arc as a function of velocity and stimulus type. Thresholds were obtained at a 60° listening azimuth. Standard deviations are given in parentheses. For MAAs (column 1), thresholds were interpolated (or extrapolated^a) from fitted psychometric functions, and the standard errors of the estimate (when available) are given in parentheses. Estimated minimum integration times are given in the last column. See text.

	0 (MAA)	10	20	Velocity (°/s) 45	90	180	Minimum integration time (ms)
Subject: TT							
3-kHz tone	13.2 (8%)	14.4 (3.0)	26.5 (1.3)	24.5 (6.1)	33.2 (5.5)	48.0 (12.9)	856
Third octave	6.7 (5%)	13.4 (1.2)	21.3 (5.4)	33.7 (1.1)	42.0 (0.6)	72.2 (2.8)	1627
One octave	9.0 (5%)	14.7 (0.3)	21.6 (2.5)	25.2 (4.1)	31.6 (1.9)	33.3 (9.7)	476
Wide band	1.2 ^a (2%)	7.7 (1.1)	10.1 (1.3)	15.9 (2.0)	18.6 (4.0)	26.7 (7.6)	526
Subject: MW							
3-kHz tone	5.7 (10%)	9.2 (3.8)	18.6 (2.6)	15.4 (8.5)	46.2 (8.7)	69.6 (14.7)	1788
Third octave	8.2 (11%)	11.4 (1.7)	22.0 (4.5)	25.3 (8.0)	41.8 (20.3)	56.9 (27.3)	1258
One octave	12.7 (9%)	13.7 (2.0)	17.4 (3.6)	24.1 (2.0)	30.9 (2.0)	47.2 (6.3)	949
Wide band	2.2 ^a (6%)	9.2 (2.8)	8.8 (0.6)	15.3 (2.3)	20.9 (5.6)	26.4 (1.8)	528
Subject: AG							
3-kHz tone	—	13.5 (1.4)	27.0 (5.9)	16.9 (3.2)	35.0 (5.3)	38.7 (8.8)	652
Third octave	—	11.9 (1.3)	16.9 (0.2)	40.6 (2.8)	52.5 (5.6)	79.7 (0.8)	1937
One octave	—	14.0 (3.4)	20.0 (1.8)	26.6 (2.3)	37.6 (7.2)	56.2 (10.0)	1192
Wide band	—	10.7 (0.9)	14.3 (0.7)	21.4 (5.1)	22.9 (2.8)	32.9 (6.6)	595
Subject: AC							
3-kHz tone	15.0 (18%)	13.8 (0.4)	20.0 (3.2)	20.0 (2.9)	22.8 (2.8)	51.9 (14.1)	1032
Third octave	11.3 (15%)	13.5 (1.0)	19.9 (3.8)	30.4 (9.7)	31.2 (15.9)	51.2 (6.2)	998
One octave	11.3 (4%)	11.7 (0.7)	19.7 (3.9)	21.9 (1.4)	30.0 (4.7)	35.9 (4.1)	633
Wide band	1.0 ^b	6.6 (0.4)	8.7 (1.7)	12.6 (1.5)	19.4 (4.4)	25.6 (6.2)	554
Mean across subjects							
3-kHz tone	11.3 (4.9)	12.7 (2.4)	23.0 (4.3)	19.2 (4.0)	34.3 (9.6)	52.1 (12.9)	1082 (496)
Third octave	8.7 (2.3)	12.6 (1.1)	20.0 (2.3)	32.5 (6.4)	41.9 (8.7)	65.0 (13.2)	1455 (412)
One octave	11.0 (1.9)	13.5 (1.3)	19.7 (1.7)	24.5 (2.0)	32.5 (3.5)	43.1 (10.6)	812 (320)
Wide band	1.5 (0.6)	8.3 (2.1)	11.0 (2.9)	16.6 (4.4)	20.3 (2.3)	28.4 (3.9)	551 (32)

^a Extrapolated.

^b See footnote 6.

The pattern of MAAs was quite similar to that observed with the low-velocity MAMAs, demonstrating that MAMAs in the “asymptotic region” (i.e., for sufficiently long durations) and MAAs depended similarly on bandwidth at the 60° listening azimuth as well as at midline.

As noted, high-velocity targets presented at 60° azimuth depended much more strongly on bandwidth than the low-velocity targets or MAAs. We feel that this interaction may be due to the fact that the target’s onset in the high-velocity conditions was occurring at an azimuth well beyond 60° (recall, the trajectory’s end point was programmed to occur at

an azimuth of 60° to the subject’s left, and the targets were moving left to right). Since for a 3000-Hz stimulus the azimuthal region between 60° and 90° (and that between 90° and 120°) is quite undifferentiated in terms of interaural information (Shaw, 1974), an even larger trajectory than otherwise expected would be required to make the discrimination. However, the region between 60° and 90° azimuth would become more resolvable as bandwidth increased, thus leading to better target onset localization and more normal velocity functions for the broadband noise condition.

Although this analysis would explain the steeper band-

width functions for higher-velocity than for lower-velocity targets, it does not explain the finding that high-velocity MAMAs for third-octave noises were higher than those for pure tones. One potential explanation for this result is that monaural intensity changes (in the near ear) play a significant role in detecting the movement of 3000-Hz targets presented from 60° (and beyond): such intensity information may be more useful with the (nonfluctuating) pure-tone stimulus than with the (fluctuating) narrow-band noise stimulus.

3. Estimates of MIT

Estimates of MITs were made from the velocity functions as described in Sec. 1 D 3, and are listed in the last columns of Table I (0° azimuth) and Table II (60° azimuth). For MAMAs at 0° azimuth, the estimates were not much different from each other, nor from those obtained in experiment 1: Averages across the four subjects were 335 (from experiment 1), 420, 352, and 362 ms for the pure-tone, third-octave, one-octave, and wideband noise conditions, respectively. These estimates indicate that, although auditory spatial resolution improved with stimulus bandwidth, the minimum time required to achieve optimal performance did not appear to depend systematically on bandwidth. In other words, performance deteriorated with velocity in a parallel manner for each bandwidth condition tested.

At 60° azimuth, the estimates of MIT were considerably larger than those at 0°: Averaged across subjects the estimates were 1082, 1455, 812, and 551 ms, for the pure-tone, third-octave, one-octave, and wideband conditions, respectively. These larger estimates, particularly for the first three conditions, reflect the bandwidth-by-velocity interaction discussed in the previous section. It is also notable that the individual differences in the minimum integration time estimates were quite large for the narrow-band targets (see Table II), suggesting that differences in pinna shape may have been important in this task.

V. GENERAL DISCUSSION

The results from experiments 1–3 can be summarized very simply: Auditory spatial resolution shows the same dependencies on stimulus frequency, bandwidth, and listening azimuth, over the ranges we tested, whether measured with dynamic (MAMA) or stationary (MAA) stimuli. For all conditions we tested, the MAAs were smaller than the smallest measured MAMAs, and appeared to represent a limit to which the MAMA asymptoted as stimulus velocity decreased.⁷ In most cases, the MAA was smaller than the y intercept of the MAMA-by-velocity linear regression, indicating that the interpretation of the MAA as a “zero-velocity MAMA” may be an oversimplification.

Our data support and extend the findings of Perrott and Tucker (1988), who were the first to point out the parallel dependence of MAMAs and MAAs on stimulus frequency. However, while the data from Perrott and Tucker (1988) and from the current experiments support the notion that the MAA represents the optimum measure of auditory spatial resolution, the data reported by Perrott and Marlborough

(1989) appear at first glance to be inconsistent with this view. These investigators reported that the MAMA measured with a target that stayed on during its trajectory was smaller than the “MAA” measured when a moving loudspeaker was activated only at its end points; they concluded that the moving targets must have been processed differently from stationary targets.

The “MAA” as measured by Perrott and Marlborough (1989) is not a typical index of static auditory spatial resolution, since the markers employed to delineate spatial positions were not themselves static; this fact, when considered along with the very brief durations of the markers (10 ms), makes it likely that Perrott and Marlborough’s measure of static spatial resolution did not reflect subjects’ optimum resolution ability under static conditions. Had such optimum measurements been made, we would predict that they would have been at least as small as their subjects’ best MAMAs.

On the other hand, Perrott and Marlborough (1989) obtained quite low MAMAs (1.0° for a wideband target moving 20°/s). In our experiment 3, we were unable to replicate this extremely good dynamic resolution (our average MAMA for a wideband noise with a velocity of 20°/s was 5.7°). Two notable differences between the two studies may account for the difference in measured dynamic spatial resolution. First, the task required of the subject was different: In Perrott and Marlborough’s experiment, the target moved either leftward or rightward on every trial and subjects had to respond in which direction it moved; in the present study the target was either moving or stationary. A “left”–“right” task has previously been shown to yield lower thresholds, by about a factor of 2, than a “moving”–“stationary” task (Grantham, 1985). A second difference is that the sources were approximately 80% farther away in the present study than in Perrott and Marlborough’s study (1.6 vs 0.9 m). Possibly, MAMAs measured in the near field would be smaller than those measured at farther distances. In any case, we do not believe that Perrott and Marlborough’s results are inconsistent with the notion that the horizontal-plane MAA measured under *optimum* conditions represents an upper limit of spatial resolution that the MAMA might approach, but cannot surpass.

It is noteworthy that the deterioration of spatial resolution with decreasing stimulus duration (or increasing velocity) is not limited to dynamic stimuli. Grantham (1985) and Perrott and Pacheco (1989) have shown that the MAA also deteriorates as duration (interstimulus interval) decreases. In fact, Grantham’s (1985) experiment with 500-Hz tones demonstrated that MAMAs and MAAs were virtually identical for conditions in which the total presentation time was the same for the two tasks; furthermore, performance deteriorated identically in the two tasks, as presentation time decreased down to about 100 ms. For presentation times less than 100 ms, MAMAs were about twice as large as the MAA.

VI. SUMMARY AND CONCLUSIONS

We measured MAMAs and MAAs as a function of stimulus frequency, bandwidth, and listening azimuth. Both

MAMAs and MAAs showed the following effects: (1) thresholds were higher at 3000 Hz than at frequencies either lower or higher; (2) thresholds generally decreased with bandwidth (at a 3000-Hz center frequency); and (3) thresholds were 1.5 to 3.0 times higher at a 60° listening azimuth than at midline (0°). In addition, we found no effect of the direction of motion on MAMAs measured with pure tones.

These results indicate a very close relationship between dynamic and static spatial resolution. In all conditions that we tested, the MAAs defined an upper limit of performance (a lower limit on threshold) toward which the MAMAs asymptoted as stimulus velocity was decreased. In no case was an MAMA smaller than an MAA determined under the same stimulus conditions.

In all conditions, MAMAs increased monotonically with target velocity. The rate at which MAMAs increased with velocity was used in the present experiments to estimate a minimum integration time (MIT); that is, the time required for the system to perform optimally (i.e., approach maximum static resolution capability) in a dynamic resolution task. Estimates of this minimum time for targets presented at 0° azimuth were very consistent, with the average time across subjects ranging from 292–420 ms for all frequencies and bandwidths tested.⁸ The estimates at 60° azimuth were considerably larger (average time across subjects: 812–1455 ms for the three narrowest bandwidths), and probably reflect subjects' difficulty in discriminating the positions of a narrow-band target in the azimuthal region between 60° and 120°, where the target began its trajectory. The wide-band target enabled subjects to better discriminate positions in this azimuthal region, resulting in estimated MITs (average: 512 ms) almost as good as those estimated for the 0°-azimuth conditions.

Although a quantitative connection has yet to be described, the relatively large estimates of MITs in the MAMA tasks are consistent with the description of the binaural processor as a sluggish system (Blauert, 1968, 1972; Grantham and Wightman, 1979; Kollmeier and Gilkey, 1990). For moving targets, the binaural sluggishness is manifested in the long presentation time required for performance with a moving target to reach asymptotic levels; for static targets (the MAA task) the sluggishness is manifested in a similar way in the increasingly poor response to a "step" change in position as interstimulus interval is decreased. It remains to be seen whether minimum integration times for resolution tasks that do not depend on binaural processing (e.g., in the vertical plane) will be smaller than those measured here.

ACKNOWLEDGMENTS

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¹ In addition to the random attenuation applied to all stimuli prior to presentation, 1- to 3-dB attenuation was applied to stimuli presented from the

two "boom" loudspeakers to compensate for the measured level differences due to their slightly nearer positions to the subject.

² In the most extreme case (for the 10°/s condition) the intertrial interval was 17 s, and a threshold run lasted 10–15 min.

³ We programmed the offset rather than the midpoint of the moving stimulus to occur within $\pm 1.8^\circ$ of midline because of subjects' bias to localize moving sounds in the direction of their movement (Perrott and Musicant, 1977, 1981). In earlier pilot studies, we had programmed the moving target's midpoint to occur at midline and found that subjects could reliably discriminate the higher velocity (i.e., shorter duration) targets from stationary targets, not by detecting motion, but by discriminating the "overall" positions of the two stimuli: The moving target appeared farther to the right (its direction of travel) than the stationary target, even though their physical centroids were in the same place. By fixing the average position of the moving target's end point near midline (and by programming $\pm 1.8^\circ$ of jitter), we were able to eliminate the usefulness of any localization cues, at least for the faster-velocity targets. For the slower-velocity targets, where threshold durations could reach 1000 ms or more, it is possible that subjects in some cases could perform the task by discriminating the moving and stationary targets' onset positions (the moving target's onset would be farther to the left than the stationary target's onset position). However, based on our pilot work, any such position cues are minimized when the moving target's offset position (as opposed to its onset or midpoint position) occurs at midline. To the extent there was a useful position cue at the slower velocities, our measured thresholds would be underestimates of the true MAMAs.

⁴ Starting durations and stepsizes were chosen such that the initial presentation was well above threshold and about 4–5 steps covered the range of the psychometric function from chance to perfect performance. Optimum values depended on stimulus velocity, and to a lesser extent on stimulus frequency. For example, for the 1000-Hz stimulus, typical values of starting duration/stepsize duration (in ms) for the different velocities were 800/50 (10°/s); 800/50 (20°/s); 600/50 (45°/s); 300/33 (90°/s); 300/16 (180°/s).

⁵ It can be shown that the MIT estimated in this manner is 5.0 times the slope of the linear fit of MAMA versus velocity [i.e., $5.0 \times T$ in Eqs. (1)–(3)].

⁶ For cases in which threshold estimates could not be made from the psychometric functions by interpolating on the best-fitting Gaussian ogive, an attempt was first made to extrapolate from this best-fitting function. If this estimate was unrealistic (we considered any estimate less than 1° of arc unrealistic), we employed the following procedure: First, we computed the average slope of the best-fitting psychometric functions for that subject across conditions similar to that being estimated that had resulted in realistic estimates (e.g., across tone conditions in the case of an estimate for a tone; noise conditions in the case of an estimate for noise). Fitting this average psychometric function to a single data point (performance level at the smallest angle employed) allowed a second estimate of threshold. If this second estimate was also less than 1.0°, we arbitrarily estimated the MAA as 1.0°.

⁷ Saberi and Perrott (1990) have recently shown that the MAMA does not continue to decrease as velocity continues to decrease. Rather the MAMA increases again for extremely slow velocities, describing a U-shaped function of velocity. The authors attribute this loss in resolution at the slowest velocities to effects of memory failure due to the long travel time (sometimes more than 10 s) required for the slowly moving target.

⁸ These estimates are somewhat larger than estimates made by Grantham (1986) (150–300 ms). However, the latter estimates were made by simply "eyeballing" the MAMA-by-duration functions, while those in the present experiments were derived from a least-squares fit to the data, and were based on the arbitrary decision that the function has "almost" reached its asymptote when its value is 25% greater than the asymptote.

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